

PLANNER DECISION VIEW

Release the polished-greedy plan

Exact recourse completed · independent validation passed · quality escalation available

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Decision summary

Planning metric	Default routing	Recommended plan	Change
Objective capture	61.39%	64.90%	+3.51 pp
Case fill rate	64.21%	68.50%	+4.29 pp
Reassigned orders	0	4	+4
Runtime	0.30 s	2.86 s	+2.56 s
Validation violations	0	0	No change

Objective capture is fulfilled value minus penalties and shipping, divided by total requested merchandise value. **Case fill** is fulfilled cases divided by requested cases. **pp** means percentage points.

Planner actions

1. Review the four reassignments for lane authorization and service rationale.
2. Confirm calendar exceptions and any capacity event after the data snapshot.
3. Release the plan when the candidate DCs are authorized.
4. Trigger exact LNS for material conflicts or a failed approval threshold; re-optimize after material inventory changes.

What the solver guarantees

- exact assignment, load cohesion, case balance, and integer quantities;
- candidate eligibility, protected ATP, capacity, and diversion thresholds; and
- independent objective recomputation plus rejection of invalid or non-improving moves.

Escalation guide

Signal	Action
Routine window; recommended plan passes review	Release polished-greedy plan
Coupled inventory/dock conflict or high-value exception	Run exact LNS with a fixed time budget
Small disputed subset needs a certificate	Run full exact MILP and inspect the bound
Quantum experiment requested	Use only an approved bounded or synthetic neighborhood; retain exact recourse and validation

Known limitations

This is one challenge snapshot. DC authorization, authoritative calendars, throughput maxima, customer service rules, and commercial approvals require owner confirmation. Quantum hardware remains experimental and has not shown a speed or quality advantage.

Recommended decision: approve the polished-greedy plan subject to planner review, retain exact LNS as the quality escalation, and treat quantum runs as controlled R&D.